

DB schema

Entity Relationship Diagram — Database Design Document

Karnataka Police Department

Color Legend

PK — Primary Key	Uniquely identifies each record in the table
FK — Foreign Key	References the Primary Key of another table
Alternate row	Alternating row shading for readability
Normal column	Regular data column with no key constraint

Table Definitions

CaseMaster

Column Name	Type	Key	Description
CaseMasterID	INT	PK	Primary key — unique identifier for each FIR/case
CrimeNo	VARCHAR		Crime Number is assigned at the police station level and is linked to the corresponding PoliceStationID. The Crime Number follows a structured format consisting of: 1 digit Case Category Code + 4 digit District ID + 4 digit Police Station ID (Unit ID) + 4 digit Year + 5 digit Running Serial Number A separate running serial number is maintained for each police station, case category, and year. Examples: - FIR: 104430006202600001 - UDR: 304430006202600001 - Zero FIR: 804430006202600001 - PAR: 404430006202600001
CaseNo	VARCHAR		The Case Number is generated at the police station level and is associated with the corresponding PoliceStationID. For each case category, a unique serial number is maintained per police station and per year. The format is YYYY + 5-digit running serial number (e.g., 202600001). (Last 9 digits from CrimeNo)
CrimeRegisteredDate	DATE		Date when the FIR was registered
PolicePersonID	INT	FK	FK → Employee.EmployeeID — officer who registered the FIR
PoliceStationID	INT	FK	FK → Unit.UnitID — police station where FIR is registered
CaseCategoryID	INT	FK	FK → CaseCategory.CaseCategoryID — category
GravityOffenceID	INT	FK	FK → GravityOffence.GravityOffenceID — gravity level of the offence
CrimeMajorHeadID	INT	FK	FK → CrimeHead.CrimeHeadID — major crime head classification
CrimeMinorHeadID	INT	FK	FK → CrimeSubHead.CrimeSubHeadID — minor crime sub-head classification
CaseStatusID	INT	FK	FK → CaseStatusMaster.CaseStatusID — current status of the case
CourtID	INT	FK	FK → Court.CourtID — court where the case is being heard

Column Name	Type	Key	Description
IncidentFromDate	DATETIME		Start date and time of the incident
IncidentToDate	DATETIME		End date and time of the incident
InfoReceivedPSDate	DATETIME		Date and time when police station received information about the incident
latitude	DECIMAL		GPS latitude coordinate of the incident location
longitude	DECIMAL		GPS longitude coordinate of the incident location
BriefFacts	Nvarchar(Max)		Summary of the case

ComplainantDetails

Column Name	Type	Key	Description
ComplainantID	INT	PK	Primary key — unique identifier for the complainant
CaseMasterID	INT	FK	FK → CaseMaster.CaseMasterID — FIR/case filed by this complainant
ComplainantName	VARCHAR		Full name of the complainant
AgeYear	INT		Age of the complainant
OccupationID	INT	FK	FK → OccupationMaster.OccupationID — occupation of the complainant
ReligionID	INT	FK	FK → ReligionMaster.ReligionID — religion of the complainant
CasteID	INT	FK	FK → CasteMaster.caste_master_id — caste of the complainant
GenderID	INT		Gender of the complainant (lookup value)

ActSectionAssociation

Column Name	Type	Key	Description
CaseMasterID	INT	FK	FK → CaseMaster.CaseMasterID — FIR/case this act-section applies to
ActID	INT	FK	FK → Act.ActCode — legal act under which charges are framed
SectionID	INT	FK	FK → Section.SectionCode — specific section of the act invoked
ActOrderID	INT		Display/print order of the act within the case
SectionOrderID	INT		Display/print order of the section under the act

Victim

Column Name	Type	Key	Description
VictimMasterID	INT	PK	Primary key — unique identifier for each victim
CaseMasterID	INT	FK	FK → CaseMaster.CaseMasterID — FIR/case this victim belongs to
VictimName	VARCHAR		Full name of the victim
AgeYear	INT		Age of the victim in years
GenderID	INT		Gender of the victim (lookup value) like m, f, t
VictimPolice	VARCHAR		If Victim is police then 1 else 0

Accused

Column Name	Type	Key	Description
AccusedMasterID	INT	PK	Primary key — unique identifier for each accused person
CaseMasterID	INT	FK	FK → CaseMaster.CaseMasterID — FIR/case this accused is linked

Column Name	Type	Key	Description
			to
AccusedName	VARCHAR		Full name of the accused
AgeYear	INT		Age of the accused
GenderID	INT		Gender of the accused mentioned as M/F/T
PersonID	VARCHAR		Accused Sorting like A1, A2, A3....

ArrestSurrender

Column Name	Type	Key	Description
ArrestSurrenderID	INT	PK	Primary key — unique identifier for each arrest/surrender event
CaseMasterID	INT	FK	FK → CaseMaster.CaseMasterID — FIR/case linked to this arrest/surrender
ArrestSurrenderTypeID	INT		Type of event: arrest or voluntary surrender (lookup value)
ArrestSurrenderDate	DATE		Date of arrest or surrender
ArrestSurrenderStateID	INT	FK	FK → State.StateID — state where arrest/surrender occurred
ArrestSurrenderDistrictID	INT	FK	FK → District.DistrictID — district where arrest/surrender occurred
PoliceStationID	INT	FK	FK → Unit.UnitID — police station handling the arrest
IOID	INT	FK	FK → Employee.EmployeeID — Investigating Officer who made the arrest
CourtID	INT	FK	FK → Court.CourtID — court before which accused was produced
AccusedMasterID	INT	FK	FK → Accused.AccusedMasterID — accused person linked to this arrest/surrender
IsAccused	BIT		Flag (0/1): whether the person is the primary accused in the case
IsComplainantAccused	BIT		Flag (0/1): whether the complainant is also listed as accused

Act

Column Name	Type	Key	Description
ActCode	VARCHAR	PK	Primary key — unique code for the legal act (e.g. IPC, NDPS)
ActDescription	VARCHAR		Full official name/description of the act
ShortName	VARCHAR		Abbreviated/common name of the act
Active	BIT		Whether the act is currently active and usable (1=Active, 0=Inactive)

Section

Column Name	Type	Key	Description
ActCode	VARCHAR	FK	FK → Act.ActCode — parent act this section belongs to
SectionCode	VARCHAR		Section number/code (e.g. 302, 307)
SectionDescription	VARCHAR		Full description of the section
Active	BIT		Whether the section is currently active (1=Active, 0=Inactive)

CrimeHeadActSection

Column Name	Type	Key	Description
CrimeHeadID	INT	FK	FK → CrimeHead.CrimeHeadID — crime head this act-section combination maps to

Column Name	Type	Key	Description
ActCode	VARCHAR	FK	FK → Act.ActCode — legal act linked to this crime head
SectionCode	VARCHAR		Section code from the act applicable to this crime head

CrimeHead

Column Name	Type	Key	Description
CrimeHeadID	INT	PK	Primary key — unique identifier for the major crime head
CrimeGroupName	VARCHAR		Name of the crime group/major head (e.g. Crimes Against Body)
Active	BIT		Whether this crime head is active (1=Active, 0=Inactive)

CrimeSubHead

Column Name	Type	Key	Description
CrimeSubHeadID	INT	PK	Primary key — unique identifier for the crime sub-head
CrimeHeadID	INT	FK	FK → CrimeHead.CrimeHeadID — parent major crime head this belongs to
CrimeHeadName	VARCHAR		Name of this crime sub-head (e.g. Murder, Robbery)
SeqID	INT		Display/sort sequence number for ordering sub-heads

CasteMaster

Column Name	Type	Key	Description
caste_master_id	INT	PK	Primary key — unique identifier for each caste. Referenced by ComplainantDetails.CasteID
caste_master_name	VARCHAR		Name of the caste

ReligionMaster

Column Name	Type	Key	Description
ReligionID	INT	PK	Primary key — unique identifier for each religion. Referenced by ComplainantDetails.ReligionID
ReligionName	VARCHAR		Name of the religion (e.g. Hindu, Muslim, Christian)

OccupationMaster

Column Name	Type	Key	Description
OccupationID	INT	PK	Primary key — unique identifier for each occupation. Referenced by ComplainantDetails.OccupationID
OccupationName	VARCHAR		Name of the occupation (e.g. Farmer, Government Employee)

CaseStatusMaster

Column Name	Type	Key	Description
CaseStatusID	INT	PK	Primary key — unique identifier for each case status. Referenced by CaseMaster.CaseStatusID
CaseStatusName	VARCHAR		Name of the status (e.g. Under Investigation, Charge Sheeted, Closed)

Court

Column Name	Type	Key	Description
CourtID	INT	PK	Primary key — unique identifier for the court. Referenced by CaseMaster.CourtID, ArrestSurrender.CourtID
CourtName	VARCHAR		Full name of the court
DistrictID	INT	FK	FK → District.DistrictID — district where the court is located
StateID	INT	FK	FK → State.StateID — state where the court is located
Active	BIT		Whether the court is active (1=Active, 0=Inactive)

District

Column Name	Type	Key	Description
DistrictID	INT	PK	Primary key — unique identifier for the district. Referenced by Court, Unit, Employee, ArrestSurrender
DistrictName	VARCHAR		Name of the district
StateID	INT	FK	FK → State.StateID — state this district belongs to
Active	BIT		Whether the district record is active (1=Active, 0=Inactive)

State

Column Name	Type	Key	Description
StateID	INT	PK	Primary key — unique identifier for the state. Referenced by Court, District, Unit, ArrestSurrender
StateName	VARCHAR		Name of the state
NationalityID	INT		Nationality reference ID
Active	BIT		Whether the state record is active (1=Active, 0=Inactive)

Unit

Column Name	Type	Key	Description
UnitID	INT	PK	Primary key — unique identifier for the police unit. Referenced by CaseMaster.PoliceStationID, Employee.UnitID, ArrestSurrender.PoliceStationID
UnitName	VARCHAR		Name of the unit or police station
TypeID	INT	FK	FK → UnitType.UnitTypeID — type/category of the unit
ParentUnit	INT		Parent unit ID for hierarchy (self-reference to UnitID)
NationalityID	INT		Nationality reference ID
StateID	INT	FK	FK → State.StateID — state the unit belongs to
DistrictID	INT	FK	FK → District.DistrictID — district the unit belongs to
Active	BIT		Whether the unit is active (1=Active, 0=Inactive)

UnitType

Column Name	Type	Key	Description
UnitTypeID	INT	PK	Primary key — unique identifier for the unit type. Referenced by Unit.TypeID
UnitTypeName	VARCHAR		Name of the unit type (e.g. Police Station, Circle Office)
CityDistState	VARCHAR		Operational level: City / District / State

Column Name	Type	Key	Description
Hierarchy	INT		Hierarchy level number (lower = higher authority)
Active	BIT		Whether the unit type is active (1=Active, 0=Inactive)

Rank

Column Name	Type	Key	Description
RankID	INT	PK	Primary key — unique identifier for the rank. Referenced by Employee.RankID
RankName	VARCHAR		Name of the police rank (e.g. Constable, Inspector, DSP)
Hierarchy	INT		Rank hierarchy level (lower = higher rank)
Active	BIT		Whether the rank is active (1=Active, 0=Inactive)

Designation

Column Name	Type	Key	Description
DesignationID	INT	PK	Primary key — unique identifier for the designation. Referenced by Employee.DesignationID
DesignationName	VARCHAR		Name of the designation (e.g. Investigating Officer, SHO)
Active	BIT		Whether the designation is active (1=Active, 0=Inactive)
SortOrder	INT		Display sort order for dropdowns/reports

Employee

Column Name	Type	Key	Description
EmployeeID	INT	PK	Primary key — unique identifier for the police employee. Referenced by CaseMaster.PolicePersonID, ArrestSurrender.IOID
DistrictID	INT	FK	FK → District.DistrictID — district the employee is currently posted in
UnitID	INT	FK	FK → Unit.UnitID — unit/police station the employee is assigned to
RankID	INT	FK	FK → Rank.RankID — current rank of the employee
DesignationID	INT	FK	FK → Designation.DesignationID — current designation of the employee
KGID	VARCHAR		Karnataka Government ID (unique government employee number)
FirstName	VARCHAR		First name of the employee
EmployeeDOB	DATE		Date of birth of the employee
GenderID	INT		Gender of the employee (lookup value)
BloodGroupID	INT		Blood group of the employee (lookup value)
PhysicallyChallenged	BIT		Flag: whether the employee is physically challenged (1=Yes, 0=No)
AppointmentDate	DATE		Date of appointment to government service

CaseCategory

Column Name	Type	Key	Description
CaseCategoryID	INT	PK	Primary key — unique identifier for the case category. Referenced by CaseMaster.CaseCategoryID

Column Name	Type	Key	Description
LookupValue	VARCHAR		Category name (FIR, UDR, PAR..)

GravityOffence

Column Name	Type	Key	Description
GravityOffenceID	INT	PK	Primary key — unique identifier for the gravity level. Referenced by CaseMaster.GravityOffenceID
LookupValue	VARCHAR		Gravity description (e.g. Heinous, Non-Heinous)

ChargesheetDetails

Column Name	Type	Key	Description
CSID	INT	PK	Primary key — unique identifier for the chargesheet
CaseMasterID	INT	FK	FK → CaseMaster.CaseMasterID — FIR/case filed by this complainant
csdate	DATETIME		Chargesheeted date
cstype	CHAR		Final report type A-> Chargesheet, B->False Case, C->Undetected
PolicePersonID	INT	FK	FK → employeeMaster.employee ID

Relationship Matrix

Defines all foreign key relationships between tables, including cardinality and a brief description.

Parent Table	Parent Column	Relationship	Child Table	Child Column	Description
CaseMaster	CaseMasterID	One to Many	Victim	CaseMasterID	One FIR can have multiple victims
CaseMaster	CaseMasterID	One to Many	Accused	CaseMasterID	One FIR can have multiple accused persons
CaseMaster	CaseMasterID	One to Many	ArrestSurrender	CaseMasterID	One FIR can have multiple arrest/surrender events
CaseMaster	CaseMasterID	One to Many	ComplainantDetails	CaseMasterID	One FIR can have multiple complainants
CaseMaster	CaseMasterID	One to Many	ActSectionAssociation	CaseMasterID	One FIR can invoke multiple act-sections
CaseMaster	CaseMasterID	One to One	Inv_OccuranceTime	CaseMasterID	One FIR has one occurrence time/location record

Parent Table	Parent Column	Relationship	Child Table	Child Column	Description
CaseMaster	CaseCategoryID	Many to One	CaseCategory	CaseCategoryID	Many FIRs can share the same category
CaseMaster	GravityOffenceID	Many to One	GravityOffence	GravityOffenceID	Many FIRs can have the same gravity level
CaseMaster	CrimeMajorHeadID	Many to One	CrimeHead	CrimeHeadID	Many FIRs can share the same major crime head
CaseMaster	CrimeMinorHeadID	Many to One	CrimeSubHead	CrimeSubHeadID	Many FIRs can share the same crime sub-head
CaseMaster	CaseStatusID	Many to One	CaseStatusMaster	CaseStatusID	Many FIRs can have the same status
CaseMaster	CourtID	Many to One	Court	CourtID	Many FIRs can be tried in the same court
CaseMaster	PolicePersonID	Many to One	Employee	EmployeeID	Many FIRs can be registered by the same employee
ArrestSurrender	AccusedMasterID(via junction)	One to Many	inv_arrestsurrenderaccused	ArrestSurrenderID	One arrest event can link multiple accused via junction
inv_arrestsurrenderaccused	ArrestSurrenderID	Many to One	ArrestSurrender	ArrestSurrenderID	Junction links to the arrest/surrender event
ArrestSurrender	ArrestSurrenderStateID	Many to One	State	StateID	Many arrest events can occur in the same state
ArrestSurrender	ArrestSurrenderDistrictID	Many to One	District	DistrictID	Many arrest events can occur in the same district
ArrestSurrender	CourtID	Many to One	Court	CourtID	Accused may be produced before a court
ArrestSurrender	IOID	Many to One	Employee	EmployeeID	Many arrests can be made by the same IO
ComplainantDetails	OccupationID	Many to One	OccupationMaster	OccupationID	Many complainants can share the same occupation
ComplainantDetails	ReligionID	Many to One	ReligionMaster	ReligionID	Many complainants can share the same religion
ComplainantDetails	CastID	Many to One	CasteMaster	caste_master_id	Many complainants can belong to

Parent Table	Parent Column	Relationship	Child Table	Child Column	Description
					the same caste
ActSectionAssociation	ActID	Many to One	Act	ActCode	Many case-sections can reference the same act
ActSectionAssociation	SectionID	Many to One	Section	SectionCode	Many cases can use the same section
CrimeSubHead	CrimeHeadID	Many to One	CrimeHead	CrimeHeadID	Multiple sub-heads fall under one major crime head
CrimeHead	CrimeHeadID	One to Many	CrimeHeadActSection	CrimeHeadID	One crime head can map to multiple act-sections
Act	ActCode	One to Many	CrimeHeadActSection	ActCode	One act can be linked to multiple crime heads
Act	ActCode	One to Many	Section	ActCode	One act contains multiple sections
Court	DistrictID	Many to One	District	DistrictID	Many courts can be in the same district
District	StateID	Many to One	State	StateID	Many districts belong to one state
Unit	TypeID	Many to One	UnitType	UnitTypeID	Many units share the same unit type
Unit	StateID	Many to One	State	StateID	Many units are located in the same state
Unit	DistrictID	Many to One	District	DistrictID	Many units belong to the same district
Employee	DistrictID	Many to One	District	DistrictID	Many employees posted in the same district
Employee	UnitID	Many to One	Unit	UnitID	Many employees assigned to the same unit
Employee	RankID	Many to One	Rank	RankID	Many employees can hold the same rank
Employee	DesignationID	Many to One	Designation	DesignationID	Many employees can have the same designation